

论文笔记: LLM-DAMVC@A Large Language Model Assisted Dynamic Agent for Multi-View Clustering

《从表示学习中的一个中心视角来看待基于图的多视图聚类》

简介信息

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论文结构

- Abstract
- 1.Introduction
- 2.Related Work
- 3.Methodology
- 4.Experiment
- 5.Conclusions



研究问题

现有多视图聚类方法主要面临两个挑战

- 现有方法往往在 **特征域提取特征**, 对噪声敏感且可能忽略聚类特定信息 (在原始空间不可察觉或不可区分)

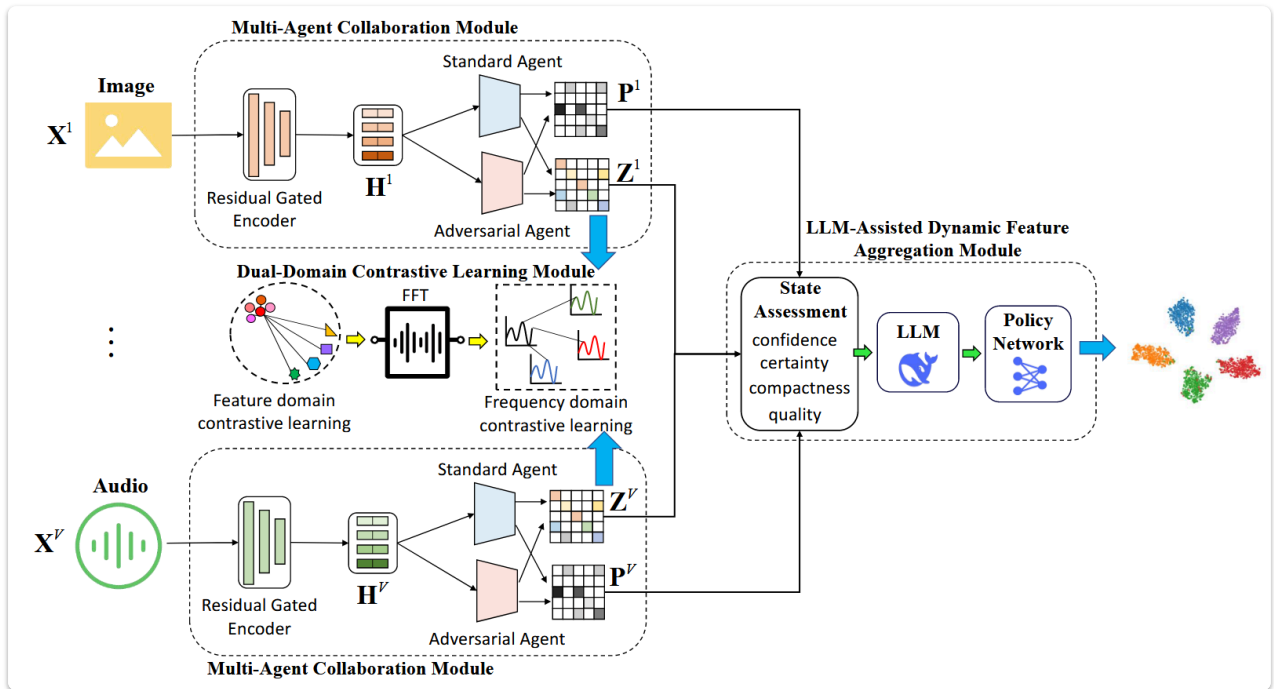
- 当前动态融合方法采用**静态策略学习权重**，缺失了复杂环境下根据视图分布和质量调整视图权重的自适应能力。换句话说，现有方法通常依赖**预定义的规则**来为每个视图分配权重，使得他们在应用于复杂场景和处理异构视图时缺乏灵活性。
- 现有方法通常缺乏能够理解视图内容、评估视图质量和执行智能决策的**集中协调机制**



研究方法

受益于大语言模型强大的决策能力，本文提出了一个新颖的用于多视图聚类的LLM辅助的动态代理（LLM-Assisted Dynamic Agent for Multi-View Clustering, LLM-DAMVC），详情如下：

- **多代理协同模块**，包含残差门控编码器和两个代理（标准化Agent和对抗Agent），用于学习跨视图统一表示。
- **双域对比学习模块**，利用特征空间对比学习增强局部辨别度，和利用频域空间对比学习增强全局结构一致性
- **LLM辅助的动态特征融合模块**，将代理的平均预测置信度、聚类结构质量、预测不确定性和表示质量分数封装送入LLMs以学习每个Agent的自适应阈值，以保留有效的Agent进行融合即可。



MULTI-AGENT COLLABORATION MODULE（多代理协同模块）

Residual Gated Encoder（残差门控网络）

正常的MLP映射

$$\mathbf{H}_l^v = \text{ReLU} \left(\text{BN}(\mathbf{H}_{l-1}^v \mathbf{W}_l + \mathbf{b}_l) \right), \quad l = 1, 2, \dots, L - 1$$

最后一层表示采用自门控机制+残差连接来增强表示

$$\mathbf{H}^v = (1 + \text{sigmoid}(\mathbf{H}_{L-1}^v \mathbf{W}_{\text{gate}})) \odot \mathbf{H}_{L-1}^v + \alpha f(\mathbf{X}^v)$$

Standard Agent (标准化代理)

共享的聚类层 (包含两层线性层、一层BN层、一层ReLU激活函数和一层Softmax)

a standard embedding. To be specific, we utilize a learnable mapping ϕ_{std} composed of two linear layers, a BN layer, a ReLU activation layer, and a softmax operation to learn the clustering probability distribution $\mathbf{P}_{\text{std}}^v \in \mathbb{R}^{N \times K}$ from $\mathbf{H}^v$ as follows:

$$\mathbf{P}_{\text{std}}^v = \phi_{\text{std}}(\mathbf{H}^v) \quad (3)$$

共享的MLP层 (包含两层线性层、一层BN层和一层ReLU激活函数)

Then we employ another learnable mapping $\mathcal{E}_{\text{std}}$ composed of two linear layers, a BN layer, and a ReLU activation layer, to project $\mathbf{H}^v$ into a standard embedding $\mathbf{Z}_{\text{std}}^v \in \mathbb{R}^{N \times d_z}$:

$$\mathbf{Z}_{\text{std}}^v = \mathcal{E}_{\text{std}}(\mathbf{H}^v) \quad (4)$$

损失约束跨视图一致性

$$\mathcal{L}_{\text{align}} = - \sum_{v=1}^V \sum_{u \neq v} \mathcal{A}(\mathbf{H}^v, \mathbf{H}^u) + \frac{1}{V(V-1)} \sum_{v=1}^V \sum_{u \neq v} \mathcal{D}_{\text{KL}}(\mathbf{P}_{\text{std}}^v \parallel \mathbf{P}_{\text{std}}^u) \quad (9)$$

where $\mathcal{A}(\cdot, \cdot)$ maximizes the canonical correlation between view-specific latent features $\mathbf{H}^v$ by operating on their cross-covariance matrix, and $\mathcal{D}_{\text{KL}}$ is KL-divergence to ensure cluster distribution consistency.

Adversarial Agent (对抗代理)

共享的聚类层 (包含两层线性层、一层BN层、一层ReLU激活函数和一层Softmax)

aries of the feature space to enhance robustness and discriminability. It first learns the adversary clustering probability distribution $\mathbf{P}_{\text{adv}}^v$ through ϕ_{adv} that has the same structure as ϕ_{std} :

$$\mathbf{P}_{\text{adv}}^v = \phi_{\text{adv}}(\mathbf{H}^v) \quad (5)$$

共享的MLP层 (包含两层线性层、一层BN层和一层ReLU激活函数)

To obtain the adversarial embedding $\mathbf{Z}_{\text{adv}}^v \in \mathbb{R}^{N \times d_z}$, we use a mapping $\mathcal{E}_{\text{adv}}$ with similar structure to $\mathcal{E}_{\text{std}}$:

$$\mathbf{Z}_{\text{adv}}^v = \mathcal{E}_{\text{adv}}(\mathbf{H}^v) \quad (6)$$

利用预测分数构造伪标签, 引导对抗样本的生成

Then, we compute $\hat{\mathbf{y}}^v = \arg \max(\mathbf{P}_{\text{adv}}^v)$ as pseudo-labels, which guides the generation of adversarial samples $\tilde{\mathbf{X}}^v$ by computing input gradients:

$$\tilde{\mathbf{X}}^v = \mathbf{X}^v + \delta \cdot \text{sign}(\nabla_{\mathbf{X}^v} \text{CE}(\mathbf{P}_{\text{adv}}^v, \hat{\mathbf{y}}^v)) \quad (7)$$

where ∇ represents gradient operation, CE denotes the cross-entropy operation, sign is the sign function, and $\delta > 0$ controls the perturbation intensity. Then, $\tilde{\mathbf{X}}^v$ is used to retrain the adversarial agent, enforcing robustness against input perturbations, and is not propagated to subsequent modules.

构造Discriminator去增强表示的鲁棒性

generated features ($\tilde{\mathbf{H}}^v = \mathbf{E}_v(\tilde{\mathbf{X}}^v)$). This adversarial branch introduces a min-max game between the encoder and the discriminator to enhance representation robustness with discriminator loss $\mathcal{L}_{\text{disc}}$.

$$\mathcal{L}_{\text{disc}} = -\mathbb{E}_{\mathbf{X}^v} [\log(\psi(\mathbf{H}^v))] - \mathbb{E}_{\tilde{\mathbf{X}}^v} [\log(1 - (\psi(\tilde{\mathbf{H}}^v)))] \quad (8)$$

其中 $\psi(\cdot)$ 是将表示映射到一维特征的函数

DUAL-DOMAIN CONTRASTIVE LEARNING MODULE (双域对比学习)

Feature-domain Contrastive Learning (特征域对比学习)

$$\mathcal{L}_{\text{feature}} = -\frac{1}{N} \sum_{1 \leq v \leq V, u \neq v} \sum_{1 \leq i \leq N} \log \frac{\exp(s_{ii}^{v,u}/\tau)}{\sum_{j=1}^N \exp(s_{ij}^{v,u}/\tau)}$$

Frequency-domain Contrastive Learning (频域对比学习)

- 首先, 利用快速傅里叶变换将表征投影到频域空间
- 然后, 提取其振幅变量 (表征跨频率分量的能量分布, 同时保持平移不变性)
- 最后, 基于振幅表征进行对比学习

$$\mathcal{L}_{\text{frequency}} = \frac{1}{NMV} \sum_{1 \leq v \leq V, u \neq v} \sum_{1 \leq i \leq N} \sum_{j \in \mathcal{J}(i)} \frac{\| |\hat{\mathbf{z}}_i^v| - |\hat{\mathbf{z}}_i^u| \|_1}{\| |\hat{\mathbf{z}}_i^v| - |\hat{\mathbf{z}}_j^u| \|_1 + \epsilon}$$

LLM-ASSISTED DYNAMIC FEATURE AGGREGATION MODULE

构造质量指示矢量

For an agent of the v -th view with type a ($a \in \{\text{adv}, \text{std}\}$), we compute a quality indicator vector $\mathbf{q}_a^v \in \mathbb{R}^{4 \times 1}$ to provide a multi-faceted assessment of its real-time performance:

$$\mathbf{q}_a^v = \begin{bmatrix} \mathbb{E}[\max(\mathbf{P}_a^v(i, :))] \\ \zeta(\mathbf{H}^v, \mathbf{P}_a^v) \\ 1 - \mathcal{H}(\mathbf{P}_a^v) \\ \mathbb{E}[\mathbf{s}_{\text{adv}}^v(i)] \end{bmatrix} \quad (14)$$

where $\mathbb{E}[\max(\mathbf{P}_a^v(i, :))]$ measure the agent's average prediction confidence, $\zeta(\mathbf{H}^v, \mathbf{P}_a^v)$ measures cluster structure quality, ζ calculates a compactness score based on intra-cluster distances, $1 - \mathcal{H}(\mathbf{P}_a^v)$ measures prediction certainty, $\mathcal{H}$ is entropy operation, and $\mathbb{E}[\mathbf{s}_{\text{adv}}^v(i)]$ measures the average representation quality score, respectively.

将Prompt送入冻结的大模型学习一个自适应阈值

$$r_a^v = \begin{cases} \mathbb{E}[\max(\mathbf{P}_a^v(i, :))] \cdot \|\mathbf{H}^v\|_F, & \text{if } \mathbb{E}[\max(\mathbf{P}_a^v[i, :])] \geq \tau_c \\ 0, & \text{otherwise} \end{cases}$$

生成权重并融合，同时设置聚类损失约束

$$w_a^v = \frac{\exp(r_a^v)}{\sum_{v=1}^V \sum_{a \in \{\text{std}, \text{adv}\}} \exp(r_a^v)}. \quad (16)$$

The final aggregated prediction is obtained by weighted fusion of all agent predictions:

$$\mathbf{P} = \sum_{v=1}^V \sum_{a \in \{\text{std}, \text{adv}\}} w_a^v \mathbf{P}_a^v \in \mathbb{R}^{N \times K}. \quad (17)$$

The clustering loss is formulated as:

$$\mathcal{L}_{\text{clus}} = - \sum_{i \in \mathcal{I}_{\text{high}}} \sum_{k=1}^K y_{ik} \log \mathbf{P}_{ik} + \lambda_1 \sum_{k=1}^K \left| \bar{\mathbf{P}}_k - \frac{1}{K} \right| + \lambda_2 \sum_{i=1}^N (-\log \max(\mathbf{P}_i)) \quad (18)$$



实验结果

聚类结果

Method	NUS			BDGP			Handwritten			MNIST-USPS			Reuters			CCV		
	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
K-Means (41)	20.90	15.45	32.19	45.72	27.43	46.20	38.95	37.21	39.45	49.30	45.40	52.90	6.70	5.90	4.01	5.12	8.97	5.01
EE-IMVC (43)	22.29	10.35	39.02	88.00	71.76	87.76	89.30	81.07	89.30	76.00	68.04	76.48	19.05	11.39	16.38	23.37	18.22	26.46
ASR (44)	21.50	13.73	38.97	97.68	92.63	97.68	93.95	88.26	93.95	97.90	94.72	7.90	15.51	18.42	14.55	24.06	17.41	22.73
DSIMVC (46)	28.89	18.06	43.27	99.04	96.86	99.04	87.20	80.39	87.20	99.34	98.13	99.34	24.35	22.17	20.41	31.90	30.70	30.54
DCP (42)	19.43	5.45	30.72	97.04	92.43	97.04	85.75	85.05	85.75	99.02	97.29	99.02	26.66	32.74	21.04	20.04	16.61	14.22
MFLVC (24)	24.11	16.43	30.87	98.72	96.13	98.72	86.55	85.98	86.55	99.66	99.01	99.66	21.14	19.98	20.37	31.23	31.60	33.90
CVCL (18)	28.65	13.96	39.83	99.20	97.29	99.20	97.35	94.05	97.35	99.70	99.13	99.70	55.64	31.14	57.35	26.23	26.25	21.17
COMPLETER(25)	22.48	7.71	38.73	59.95	56.18	59.95	69.36	73.93	69.70	89.08	88.86	89.02	25.63	31.73	25.34	24.58	22.51	21.14
ADMC(45)	24.36	15.97	40.06	96.96	96.12	96.96	84.10	81.17	84.10	91.26	95.50	91.26	77.33	70.49	77.33	23.25	20.68	23.57
COPER(40)	25.55	11.67	35.54	89.65	73.92	75.96	87.55	77.15	89.05	99.88	99.64	99.88	53.15	31.10	51.79	28.06	26.32	24.57
Our Method	37.11	37.51	48.53	99.67	98.76	99.64	98.66	95.36	98.43	99.92	99.63	99.92	78.76	76.48	79.83	46.55	55.77	49.81

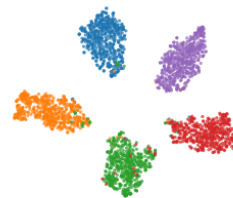
可视化结果



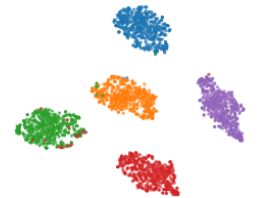
(a) Raw Features



(b) Epoch=10



(c) Epoch=50

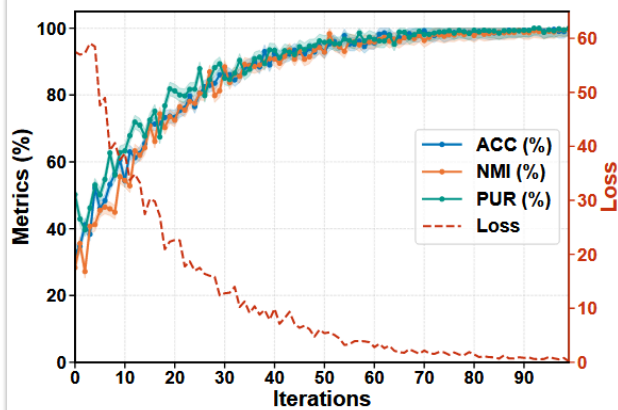


(d) Epoch=100

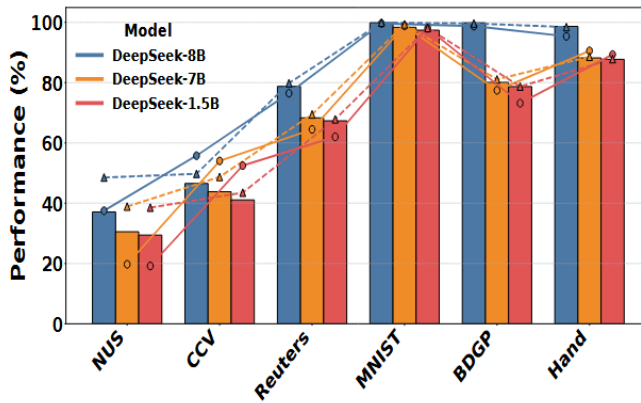
消融实验

Method	NUS			BDGP			Handwritten			MNIST-USPS			Reuters			CCV		
	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR	ACC	NMI	PUR
Baseline (Only $\mathcal{L}_{\text{feature}}$)	27.42	6.20	30.05	34.41	16.14	35.23	29.30	21.87	29.30	23.70	14.24	24.67	37.83	29.76	39.17	18.86	20.93	19.15
+ $\mathcal{L}_{\text{disc}}$	29.15	9.80	32.40	35.71	20.14	37.16	31.20	24.50	32.10	25.80	17.30	27.20	40.20	33.10	41.50	20.70	23.80	21.90
+ $\mathcal{L}_{\text{frequency}}$	31.60	14.20	36.80	40.15	26.61	40.15	55.40	48.71	55.40	65.51	62.17	65.51	45.65	38.93	47.20	23.92	28.44	25.78
+ $\mathcal{L}_{\text{clus}}$	34.37	20.44	40.67	81.73	78.15	81.73	87.75	90.30	87.75	97.30	97.73	98.30	72.82	65.48	73.20	43.61	51.14	45.53
+ $\mathcal{L}_{\text{align}}$ (Full)	37.11	37.51	48.53	99.67	98.76	99.64	98.66	95.36	98.43	99.92	99.63	99.92	78.76	76.48	79.83	46.55	55.77	49.81

收敛实验及大模型分析



(a) Convergence curve



(b) Performance with different LLMs



创新不足

- 创新
 - 首次引入LLM
- 局限
 - LLM无操作，且损失约束项过多



感悟心得

- 思考
 - 基于大模型的操作不应该这么简单，是否有以下操作
 - 将其作为Supervisor来监督或评判视图质量
 - 外部知识引导表征增强
 - 生成伪标签监督信息



笔记摘抄

- The former's impact strongly validates our core hypothesis that incorporating frequency-domain analysis is crucial for capturing global structural patterns that are often missed by conventional feature-domain methods. (前者的影响有力地验证了我们的核心假设，即结合频域分析对于捕获传统特征域方法经常遗漏的全局结构模式至关重要。)
- Existing methods generally lack a centralized coordination mechanism capable of understanding view content, assessing view quality, and performing intelligent decision-

making. (现有方法通常缺乏能够理解视图内容、评估视图质量和执行智能决策的集中协调机制。)

- Existing multi-view clustering methods primarily confront two challenges: i) they generally perform feature extraction in the feature domain, which is sensitive to noise and may neglect cluster-specific information that is indistinguishable in the original space; ii) current dynamic fusion methods adopt static strategies to learn weights, lacking capability to adjust strategies adaptively under complex scenarios according to variations in data distribution and view quality. (现有的多视图聚类方法主要面临两个挑战: i) 它们通常在特征域中进行特征提取, 该特征域对噪声敏感, 可能会忽略原始空间中无法区分的聚类特定信息; ii) 目前的动态融合方法采用静态策略来学习权重, 缺乏在复杂场景下根据数据分布和视图质量的变化自适应调整策略的能力。)